CLOUD COMPUTING

UNIT-1

PARALLEL VS DISTRIBUTED COMPUTING

Parallel vs. distributed computing

- **What is Parallel Computing?**

  In parallel computing multiple processors perform multiple tasks assigned to them simultaneously. Memory in parallel systems can either be shared or distributed. Parallel computing provides concurrency and saves time and money.

- **Examples**

  Block chains, Smartphones, Laptop computers, Internet of Things, Artificial intelligence and machine learning, Space shuttle, Supercomputers are the technologies that use Parallel computing technology.

PARALLEL VS DISTRIBUTED COMPUTING

- **What is Distributed Computing?**

  In distributed computing we have multiple autonomous computers which seems to the user as single system. In distributed systems there is no shared memory and computers communicate with each other through message passing. In distributed computing a single task is divided among different computers.

- **Examples**

  Artificial Intelligence and Machine Learning, Scientific Research and High- Performance Computing, Financial Sectors, Energy and Environment sectors, Internet of Things, Blockchain and Cryptocurrencies are the areas where distributed computing is used

PARALLEL VS. DISTRIBUTED COMPUTING

S.NO	Parallel Computing	Distributed Computing
1.	Uses single system	Uses multiple system
2.	Multiple processors perform multiple operations	Multiple computers perform single operations
3.	shared memory	Independent memory
4.	Processors communicate with each other through bus	Computers communicate with each other through message passing.
5.	Improves the system performance	Improves system scalability, fault tolerance and resource sharing capabilities
6.	tightly coupled system	Loosely coupled
7	**Speedup**: The primary aim of parallel computing is to decrease execution time	**Fault tolerance**: designed to handle partial failures
8	computation is divided among several processors sharing the same memory.	computation is divided into units and executed concurrently on different computing elements
9	Performance is focused	Performance depends on scalability, cost, reliability etc

ELEMENTS OF PARALLEL COMPUTING

What is Parallel Processing?

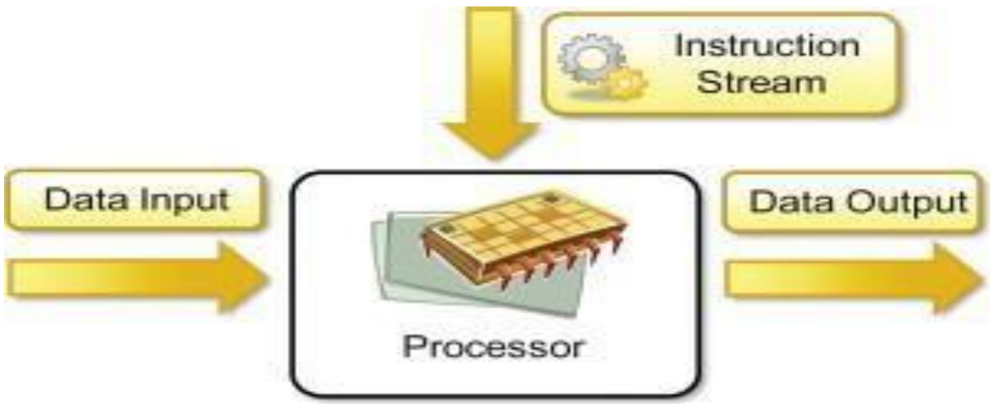
- Processing of multiple tasks simultaneously on multiple processors is called parallel processing
- A given task is divided into multiple subtasks and each one of them is processed on different CPUs.
- Programming on multi–processor system using divide-and-conquer technique is called parallel programming.
- provides a cost-effective solution

Hardware Architectures for Parallel Processing

- The core elements of parallel processing are CPUs.
- computing systems are classified into the following four categories:
  - Single Instruction Single Data (SISD)
  - Single Instruction Multiple Data (SIMD)
  - Multiple Instruction Single Data (MISD)
  - Multiple Instruction Multiple Data (MIMD)

ELEMENTS OF PARALLEL COMPUTING

- **Single Instruction Single Data (SISD)**

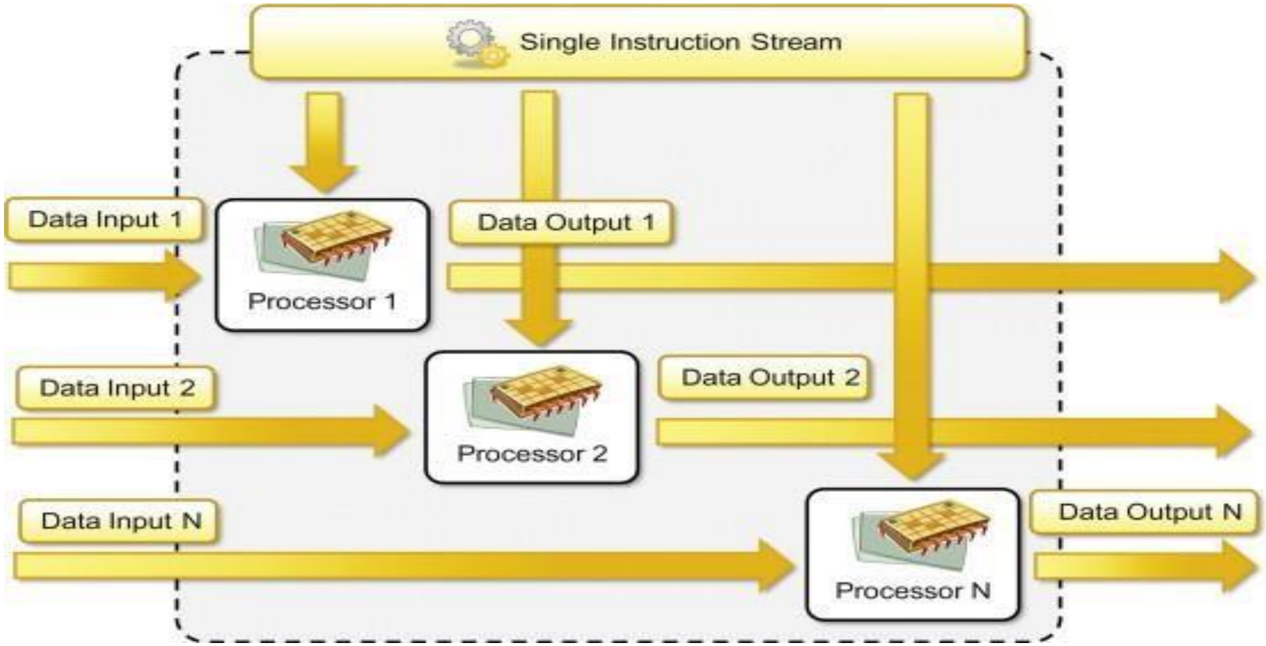


- We refer to this as a scalar processor. if it is slow it can end up dominating performance.
- uniprocessor machine executing a single instruction, which operates on a single data stream
- Instructions are processed sequentially, and hence computers adopting this model are popularly called sequential computers.
- data stored in the primary memory

ELEMENTS OF PARALLEL COMPUTING

Single Instruction Multiple Data (SIMD)

- Multiprocessor machine capable of executing the same instruction on all the CPUs, but operating on different data streams
- A single task executes simultaneously on multiple elements of data. SIMD processors are also known as array processors, since they consist of an array of functional units with a shared controller.

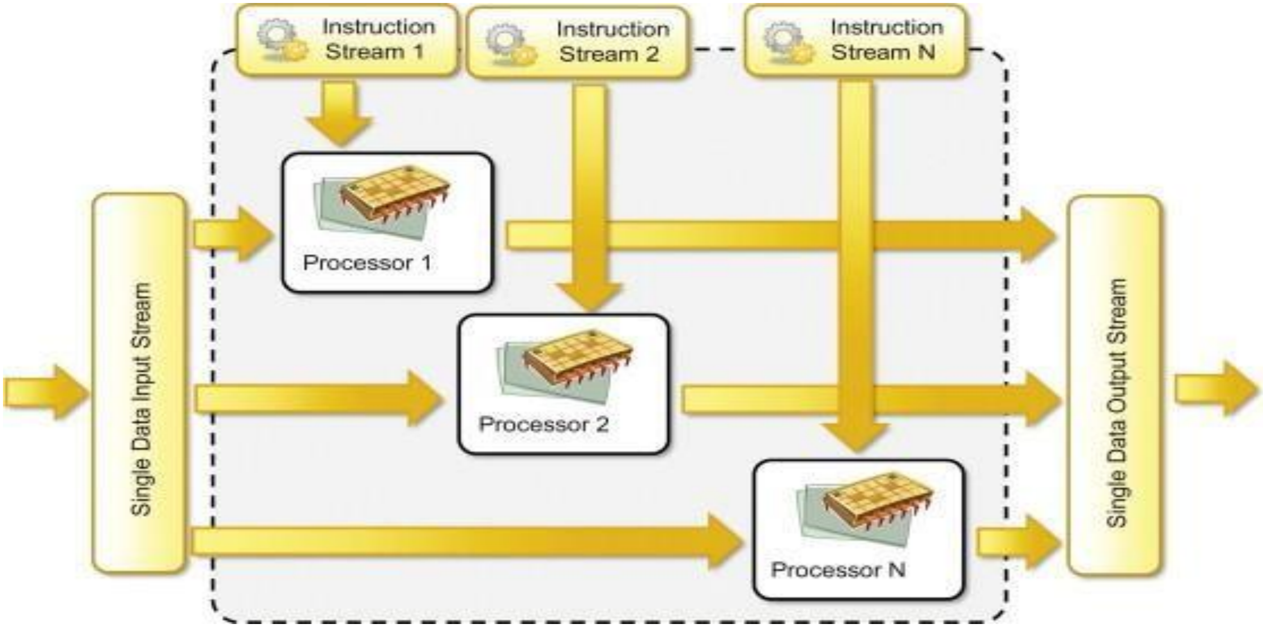


ELEMENTS OF PARALLEL COMPUTING

Multiple-instruction, single-data (MISD) systems

Multiprocessor machine capable of executing different instructions on different PEs but all of them operating on the same data set

They became more of an intellectual exercise than a practical configuration.



ELEMENTS OF PARALLEL COMPUTING

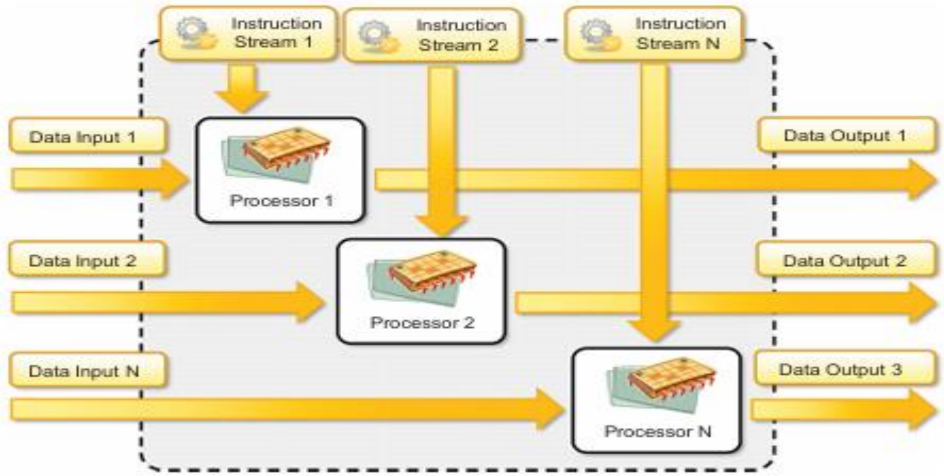
Multiple-instruction, multiple-data (MIMD) systems

Multiprocessor machine capable of executing multiple instructions on multiple data sets

Separate instruction streams, each with its own flow of control, operate on separate data. It is a heterogeneous computer.

MIMD machines are broadly categorized into two types

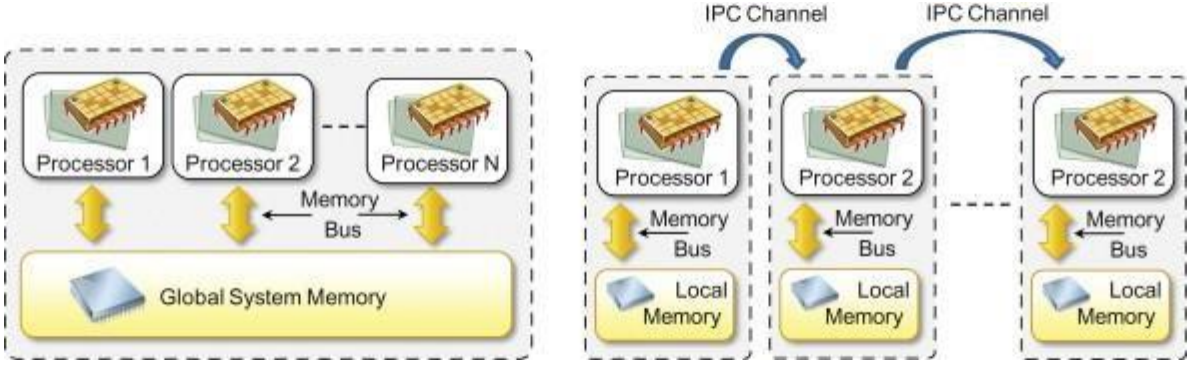
Shared-memory MIMD **Distributed-memory MIMD**



ELEMENTS OF PARALLEL COMPUTING

Shared-memory MIMD

- All the Processing Elements(PE) are connected to a single global memory, and they all have access to it Systems based on this model are also called tightly coupled multiprocessor systems
- The communication is through the shared memory
- modification of the data stored in the global memory by one PE is visible to all other PEs



Shared (left) and distributed (right) memory MIMD architecture.

ELEMENTS OF PARALLEL COMPUTING

Distributed memory MIMD machines

- All PEs have a local memory. Systems based on this model are also called loosely coupled multiprocessor systems.
- The communication between PEs in this model takes place through the interconnection network
- Easier to program but is less tolerant to failures and harder to extend with respect to the distributed memory MIMD model.
- Failures in a shared-memory MIMD affect the entire system

Approaches to Parallel Programming

- A sequential program is one which runs on a single processor and has a single line of control
- To make many processors collectively work on a single program, the program must be divided into smaller independent chunks so that each processor can work
- on separate chunks of the problem called as parallel program
- All these three models are suitable for task-level parallelism
  - Data Parallelism
  - Process Parallelism
  - Farmer and Worker Model

ELEMENTS OF PARALLEL COMPUTING

Levels of Parallelism

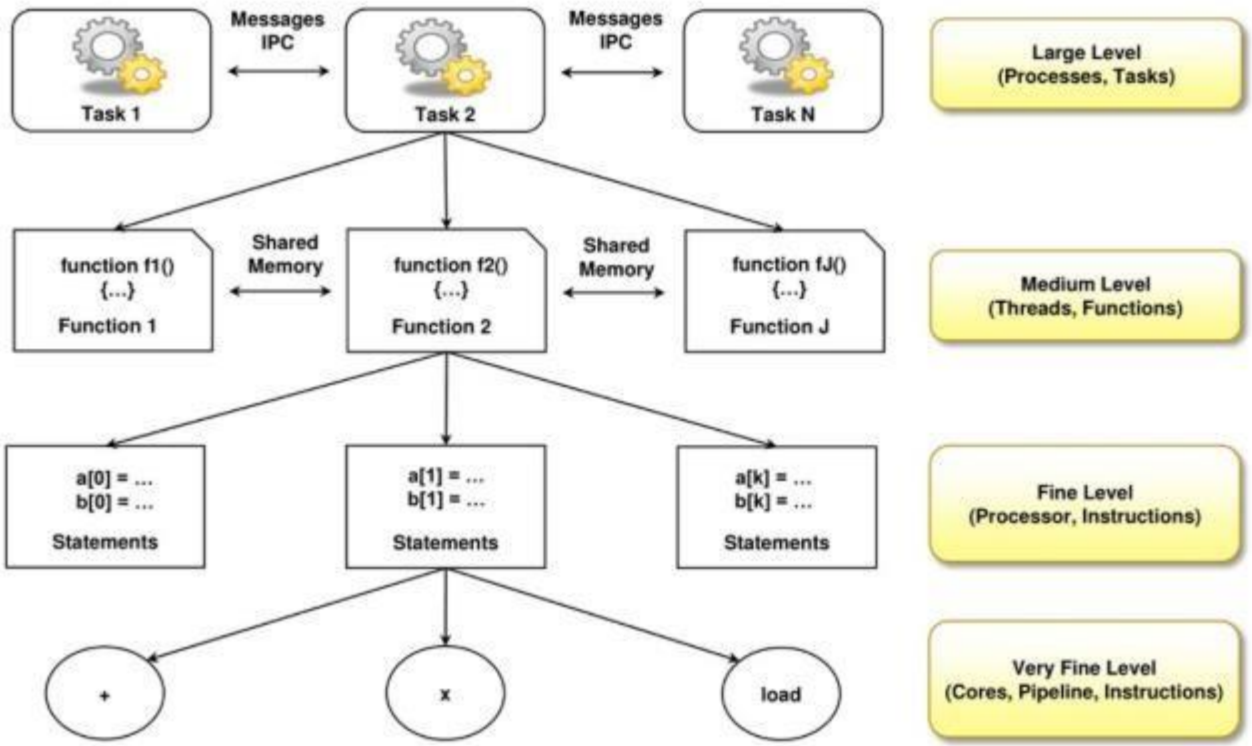
- Levels of parallelism are decided based on the lumps of code (grain size) that can be a potential candidate for parallelism
- goal to boost processor efficiency by hiding latency
- To conceal latency, there must be another thread ready to run whenever a lengthy operation occurs
- The idea is to execute concurrently two or more single-threaded applications, such as compiling, text formatting, database searching, and device simulation.

Grain Size	Code Item	Parallelized by
Large	Separate and heavy-weight process	Programmer
Medium	Function or procedure	Programmer
Fine	Loop or instruction block	Paral lelizing Compiler
Very Fine	Instruction	Processor

Parallelism within an application can be detected at several levels:

- Large-grain (or task-level)
- Medium-grain (or control-level)
- Fine-grain (data-level)
- Very-fine grain (multiple instruction issue)

ELEMENTS OF PARALLEL COMPUTING



ELEMENTS OF DISTRIBUTED COMPUTING

Here we explore how multiple activities can be performed by leveraging systems composed of multiple heterogeneous machines and systems.

General Concepts and Definitions

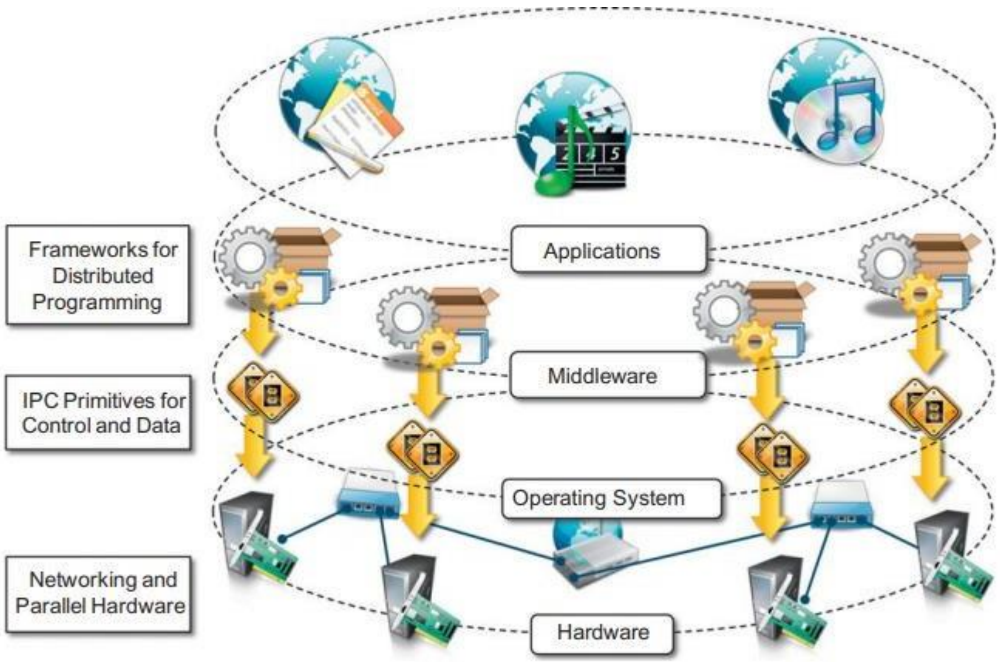
- A distributed system is a collection of independent computers that appears to its users as a single coherent system.
- A distributed system is one in which components located at networked computers
- communicate and coordinate their actions only by passing messages

Components of a distributed system

- A distributed system is the result of the interaction of several components that traverse the entire computing stack from hardware to software.
- At the very bottom layer, computer and network hardware constitute the physical infrastructure; these components are directly managed by the operating system, which provides the basic services for inter process communication (IPC)
- At the operating system level, IPC services are implemented on top of standardized communication protocols such Transmission Control Protocol/Internet Protocol (TCP/IP).

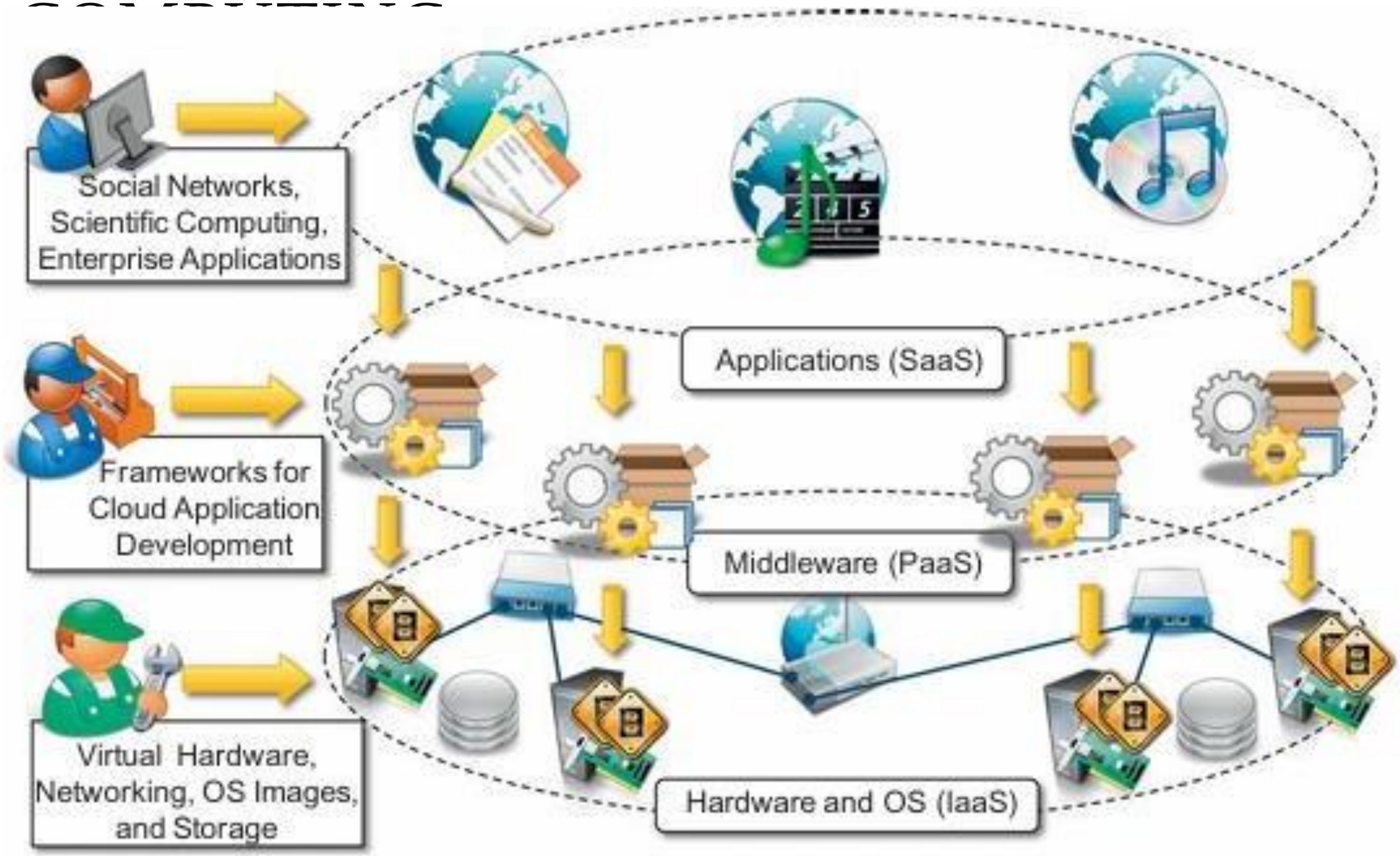
ELEMENTS OF DISTRIBUTED COMPUTING

- The middleware layer leverages such services to build a uniform environment for the development and deployment of distributed applications.
- The middleware develops its own protocols, data formats, and programming language or frameworks for the development of distributed applications.
- The top layer is represented by the applications and services designed and developed to use the middleware



layered view of a distributed system

ELEMENTS OF DISTRIBUTED COMPUTING



A cloud computing distributed system.

FUNDAMENTAL CLOUD COMPUTING

Basic Concepts and Terminology

Cloud computing is a model for enabling ubiquitous, convenient, on- demand network access to a shared pool of configurable computing resources (e.g. networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction

Cloud computing refers to both the applications delivered as services over the internet, and the hardware and system software in the datacenters that provide those services

Understanding Cloud Computing Whereas the Internet provides open access to many Web-based IT resources, a cloud is typically privately owned and offers access to IT resources that is metered.

BASIC CONCEPTS AND TERMINOLOGY

IT Resource

- An IT resource is a physical or virtual IT-related artifact that can be either software based, such as a virtual server or a custom software program, or hardware-based, such as a physical server or a network device

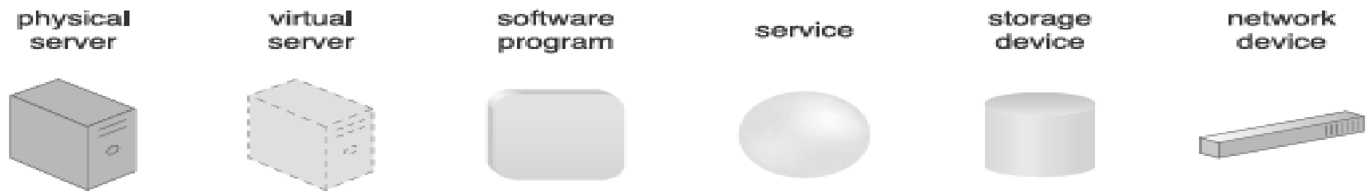


Figure 3.2
Examples of common IT resources and their corresponding symbols.

On-Premise

- An IT resource that is hosted in a conventional IT enterprise within an organizational boundary (that does not specifically represent a cloud) is considered to be located on the premises of the IT enterprise, or on-premise for short. In other words, the term “on-premise” is another way of stating “on the premises of a controlled IT environment that is not cloud-based.”

BASIC CONCEPTS AND TERMINOLOGY

Cloud Consumers and Cloud Providers

- The party that provides cloud-based IT resources is the cloud provider.
- The party that uses cloud-based IT resources is the cloud consumer.

Scaling

- Scaling, from an IT resource perspective, represents the ability of the IT resource to handle increased or decreased usage demands. The following are types of scaling:
- Horizontal Scaling – scaling out and scaling in
- Vertical Scaling – scaling up and scaling down

BASIC CONCEPTS AND TERMINOLOGY

Horizontal Scaling

- The allocating or releasing of IT resources that are of the same type is referred to as horizontal scaling .
- The horizontal allocation of resources is referred to as scaling out and the horizontal releasing of resources is referred to as scaling in.
- Horizontal scaling is a common form of scaling within cloud environments.

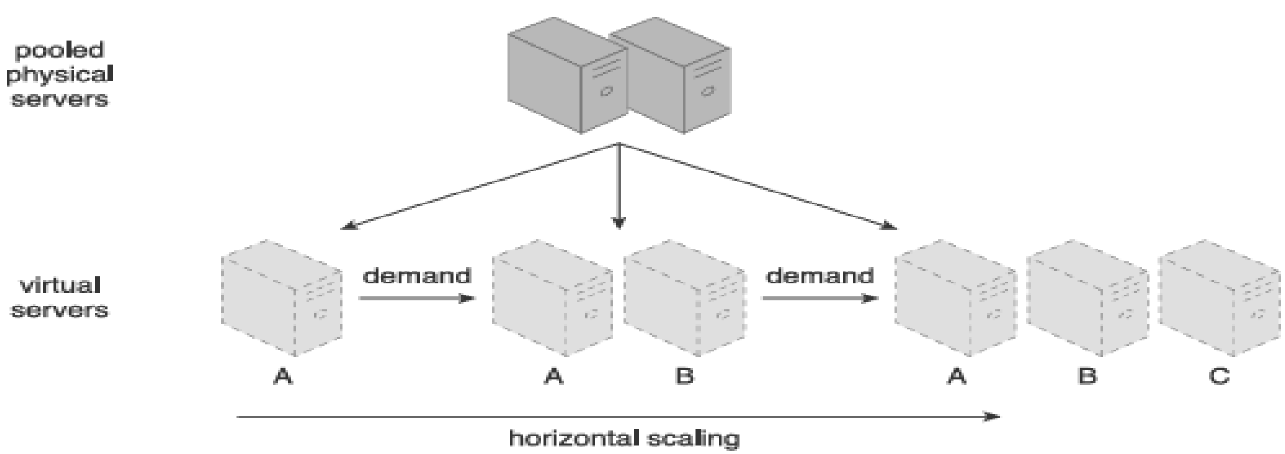


Figure 3.4

An IT resource (Virtual Server A) is scaled out by adding more of the same IT resources (Virtual Servers B and C).

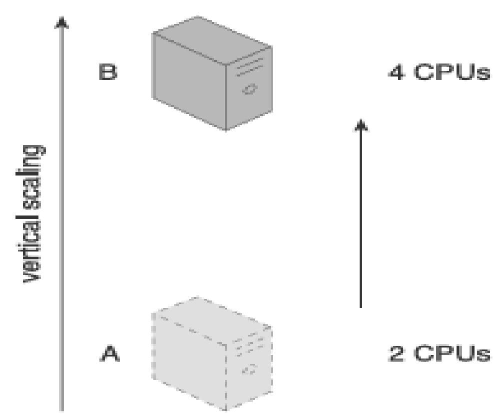
BASIC CONCEPTS AND TERMINOLOGY

Vertical Scaling

- When an existing IT resource is replaced by another with higher or lower capacity, vertical scaling is considered to have occurred. Specifically, the replacing of an IT resource with another that has a higher capacity is referred to as scaling up and the replacing an IT resource with another that has a lower capacity is considered scaling down.
- Vertical scaling is less common in cloud environments due to the downtime required while the replacement is taking place

Figure 3.5

An IT resource (a virtual server with two CPUs) is scaled up by replacing it with a more powerful IT resource with increased capacity for data storage (a physical server with four CPUs).



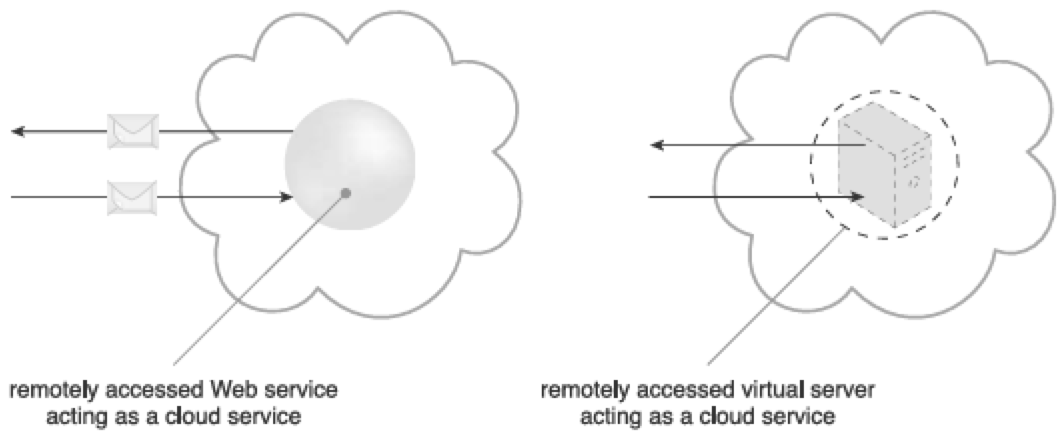
BASIC CONCEPTS AND TERMINOLOGY

Horizontal Scaling	Vertical Scaling
less expensive (through commodity hardware components)	more expensive (specialized servers)
IT resources instantly available	IT resources normally instantly available
resource replication and automated scaling	additional setup is normally needed
additional IT resources needed	no additional IT resources needed
not limited by hardware capacity	limited by maximum hardware capacity

BASIC CONCEPTS AND TERMINOLOGY

Cloud Service

- A cloud service is any IT resource that is made remotely accessible via a cloud.
- A cloud service can exist as a simple Web-based software program with a techni cal interface invoked via the use of a messaging protocol, or as a remote access point for administrative tools or larg

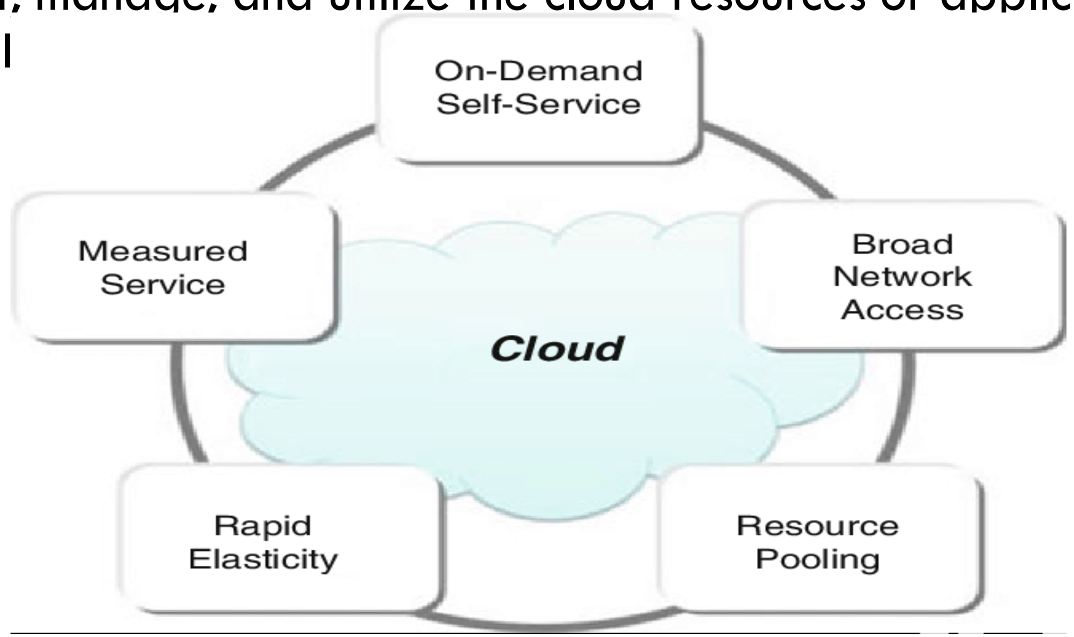


BASIC CONCEPTS AND TERMINOLOGY

Cloud Service Consumer

- A Cloud Service Consumer is the person, organization, or system that uses or accesses cloud services. They interact with cloud providers to request, manage, and utilize the cloud resources or applications remotel

Goals



GOALS AND BENEFITS

- **On-demand self-services:** The Cloud computing services does not require any human administrators, user themselves are able to provision, monitor and manage computing resources as needed.
- **Broad network access:** The Computing services are generally provided over

standard networks and heterogeneous devices.

- **Rapid elasticity:** The Computing services should have IT resources that are able to scale out and in quickly and on a need basis. Whenever the user require services it is provided to him and it is scale out as soon as its requirement gets over.
- **Resource pooling:** The IT resource (e.g., networks, servers, storage, applications, and services) present are shared across multiple applications and occupant in an uncommitted manner. Multiple clients are provided service from a same physical resource.
- **Measured service:** The resource utilization is tracked for each application and occupant, it will provide both the user and the resource provider with an account of what has been used. This is done for various reasons like monitoring billing and effective use of resource.

GOALS AND BENEFITS

Benefits of cloud computing

no upfront commitments; on demand access; nice pricing; simplified application acceleration and scalability; efficient resource allocation; energy efficiency; and seamless creation and the use of third-party services

Cost Savings for Startups

Example: Startups like Dropbox and Airbnb leveraged cloud computing to avoid significant upfront costs on hardware and software. By using cloud services, these companies managed their operations cost-effectively while scaling quickly as their user base grew.

Enhanced Collaboration for Remote Teams

Example: Companies like Slack and Zoom use cloud-based platforms to facilitate seamless collaboration among remote teams. These tools allow employees to work together in real-time, regardless of their location, improving productivity and communication.

Scalability for Retail Giants

Example: Amazon, one of the largest e-commerce platforms, uses its own AWS (Amazon Web Services) to handle massive traffic spikes during events like Black Friday. The scalability of cloud computing ensures that their systems can handle increased demand without crashing.

GOALS AND BENEFITS

Mobility

Employees have the option to compute heavy tasks from anywhere

Reduced Investments and Proportional Costs

Increased Scalability

Increased Availability and Reliability

RISKS AND CHALLENGES

Increased Security Vulnerabilities

- Shared Responsibility: Data security is managed by both the cloud provider and the consumer.
- Expanded Trust Boundaries: Consumers must trust the external cloud environment.
- Security Compatibility Issues: Public clouds may not support the same security frameworks.
- Privileged Access Risk: Providers have access to consumer data, which can be a security concern.
- Multi-Tenancy Risks: Multiple consumers using the same service can cause overlapping trust boundaries.
- Increased Threat Exposure: Shared environments can allow malicious users more chances to attack or steal data.
- Security Challenges: Hard to implement security measures that satisfy all consumers’ needs in a shared cloud setup.

RISKS AND CHALLENGES

Reduced Operational Governance Control

- Limited Control: Cloud consumers have less governance control over cloud resources compared to on premise systems.
- Operational Risks: Consumers are exposed to risks based on how the provider manages the cloud, including reliability and security.
- Example – Unreliable Provider: If a provider fails to meet its Service Level Agreements (SLAs), it can impact the quality of services used by the consumer.
- Example – Network Latency: Longer distances and extra network hops between the consumer and provider can cause latency and bandwidth issues.
- Governance Mitigation: Governance risks can be reduced using:
  - Legal contracts
  - SLAs
  - Technology audits
  - Continuous monitoring
- Cloud Governance System: Built primarily on SLAs, allowing consumers to track actual service levels and provider commitments

RISKS AND CHALLENGES

Limited Portability Between Cloud Providers

- No Universal Standards: The cloud computing industry lacks widely accepted, standardized frameworks.
- Proprietary Public Clouds: Public cloud platforms are often proprietary, meaning they use unique technologies and interfaces.
- Vendor Lock-In Risk: Custom-built consumer solutions that depend on a specific provider's platform may face difficulty migrating to other providers.
- Reduced Portability: Moving between cloud providers becomes complex and costly due to incompatible systems.

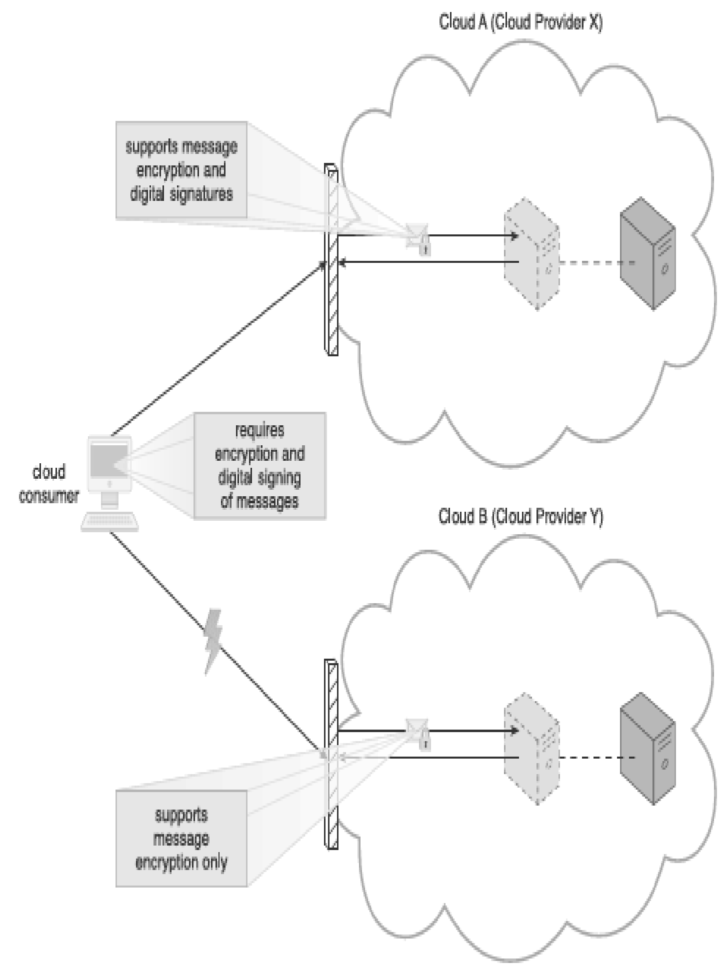


Figure 3.11

A cloud consumer's application has a decreased level of portability when assessing a potential migration from Cloud A to Cloud B, because the cloud provider of Cloud B does not support the same security technologies as Cloud A.

RISKS AND CHALLENGES

Multi-Regional Compliance and Legal Issues

- Data Residency Requirements: Some countries require certain data (e.g., personal or financial) to be stored within national borders (e.g., GDPR in EU, UK data laws).
- Uncertain Data Location: Cloud providers may store data across multiple regions or countries, often without notifying the consumer.
- Legal Jurisdiction Conflicts: Different countries have conflicting laws about data access, privacy, and disclosure, which can lead to compliance challenges.
- Government Access to Data: Some regions (e.g., U.S.) have laws (e.g., Patriot Act) that allow government access to data stored on servers within their jurisdiction—even if it belongs to foreign entities.
- Data Sovereignty Issues: Refers to the principle that data is subject to the laws of the country in which it is physically stored.
- Consumer Accountability: Despite using third-party clouds, organizations are still responsible for ensuring compliance with local and international regulations.
- Need for Transparency: Cloud consumers must request clear information from providers about data location, replication, and legal responsibilities

FUNDAMENTAL CONCEPTS AND MODELS

Roles and Boundaries

- Pre-defined Roles: Both organizations and individuals can take on specific, predefined roles in the cloud environment.
- Role-Based Interaction: Each role is defined by how it interacts with the cloud and its hosted IT resources.
- Participation in Cloud Activities: Roles are responsible for carrying out cloud-related tasks, such as accessing, managing, or maintaining services.

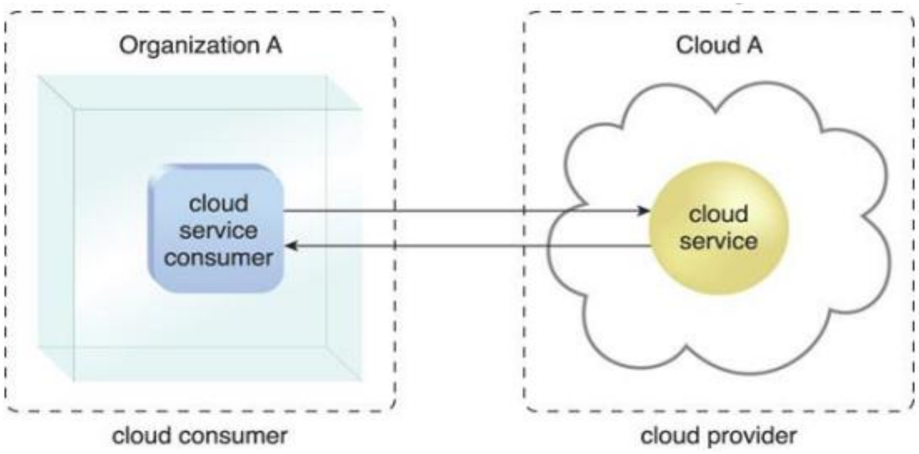
Cloud Provider

- The organization that provides cloud-based IT resources is the cloud provider. Cloud providers normally own the IT resources that are made available for lease by cloud consumers.
- some cloud providers also “resell” IT resources leased from other cloud providers.

FUNDAMENTAL CONCEPTS AND MODELS

Cloud Consumer

- A cloud consumer is an organization (or a human) that has a formal contract or arrangement with a cloud provider to use IT resources made available by the cloud provider. Specifically, the cloud consumer uses a cloud service consumer to access a cloud service



FUNDAMENTAL CONCEPTS AND MODELS

Cloud Service Owner

- The person or organization that legally owns a cloud service is called a cloud service owner. The cloud service owner can be the cloud consumer, or the cloud provider that owns the cloud within which the cloud service resides.

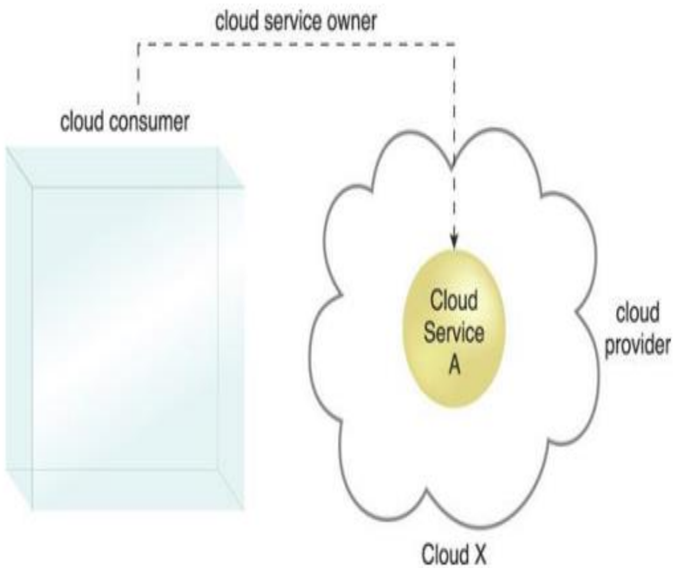


Figure 4.2. A cloud consumer can be a cloud service owner when it deploys its own service in a cloud.

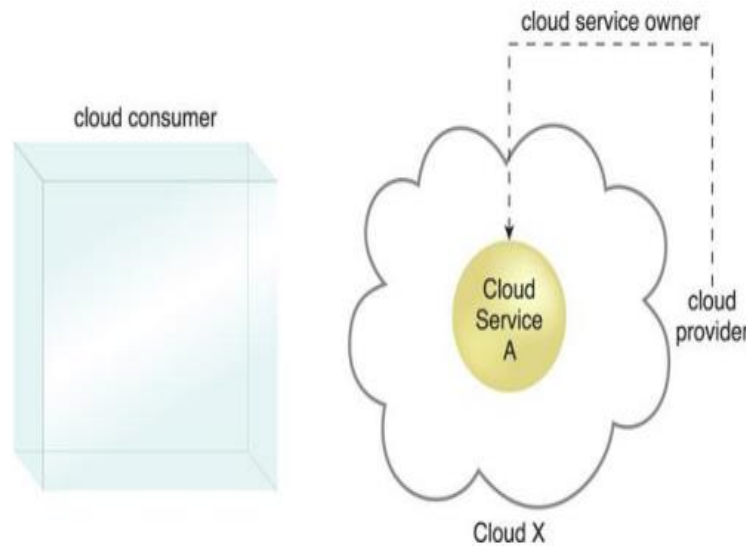


Figure 4.3. A cloud provider becomes a cloud service owner if it deploys its own cloud service, typically for other cloud consumers to use.

FUNDAMENTAL CONCEPTS AND MODELS

Cloud Resource Administrator

- A cloud resource administrator is the person or organization responsible for administering a cloud-based IT resource (including cloud services).
- The cloud resource administrator can be (or belong to) the cloud consumer or cloud provider of the cloud within which the cloud service resides.
- Alternatively, it can be (or belong to) a third-party organization contracted to administer the cloud-based IT resource.

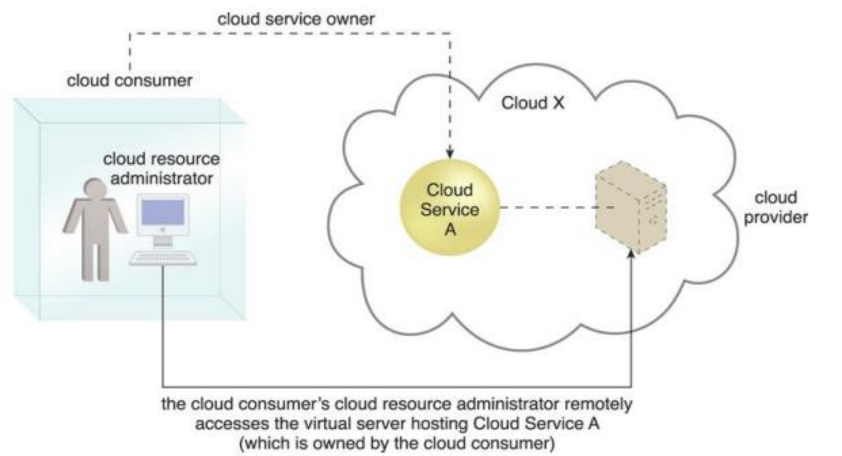


Figure 4.4. A cloud resource administrator can be with a cloud consumer organization and administer remotely accessible IT resources that belong to the cloud consumer.

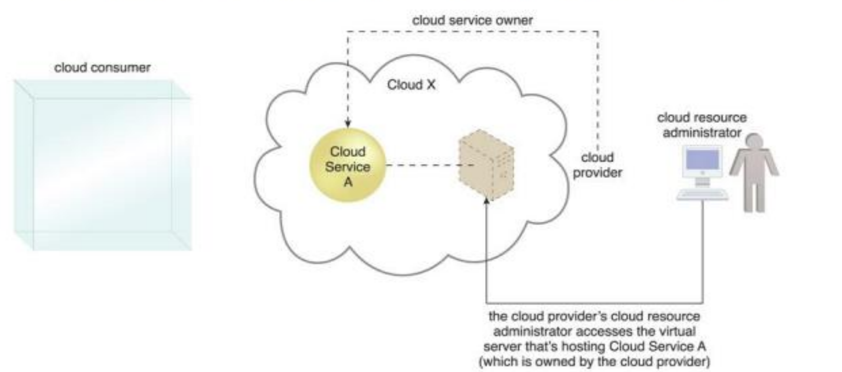


Figure 4.5. A cloud resource administrator can be with a cloud provider organization for which it can administer the cloud provider's internally and externally available IT resources.

FUNDAMENTAL CONCEPTS AND MODELS

Additional Roles

- **Cloud Auditor** – A third-party (often accredited) that conducts independent assessments of cloud environments assumes the role of the cloud auditor. The typical responsibilities associated with this role include the evaluation of security controls, privacy impacts, and performance. The main purpose of the cloud auditor role is to provide an unbiased assessment
- **Cloud Broker** – This role is assumed by a party that assumes the responsibility of managing and negotiating the usage of cloud services between cloud consumers and cloud providers
- **Cloud Carrier** – The party responsible for providing the wire-level connectivity between cloud consumers and cloud providers assumes the role of the cloud carrier. This role is often assumed by network and telecommunication providers.

FUNDAMENTAL CONCEPTS AND MODELS

Organizational Boundary

- An organizational boundary is like a fence around a company's IT resources—such as servers, data, and applications—that the organization owns and controls. It doesn't mean the entire company, just the part of its technology that it manages directly. Similarly, a cloud also has its own organizational boundary, which includes the IT resources that the cloud provider owns and manages.

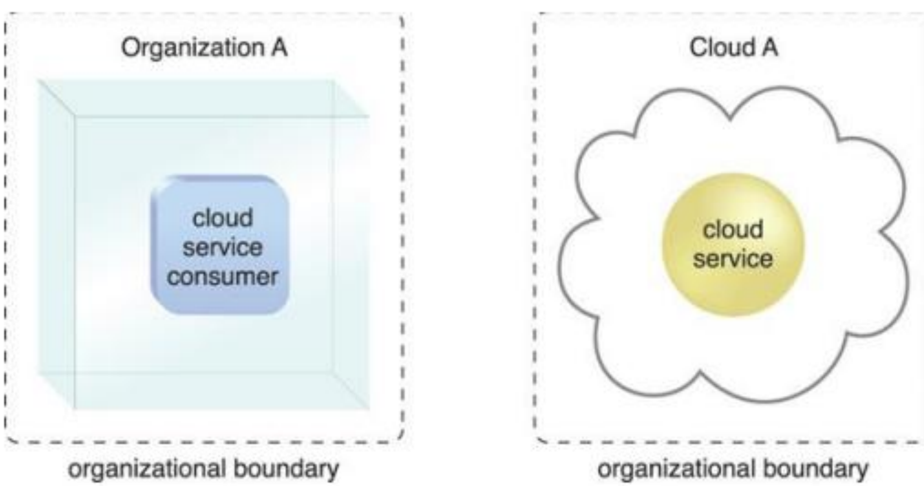


Figure 4.6. Organizational boundaries of a cloud consumer (left), and a cloud provider (right), represented by a broken line notation.

FUNDAMENTAL CONCEPTS AND MODELS

Trust Boundary

- A trust boundary is a logical perimeter that typically spans beyond physical boundaries to represent the extent to which IT resources are trusted

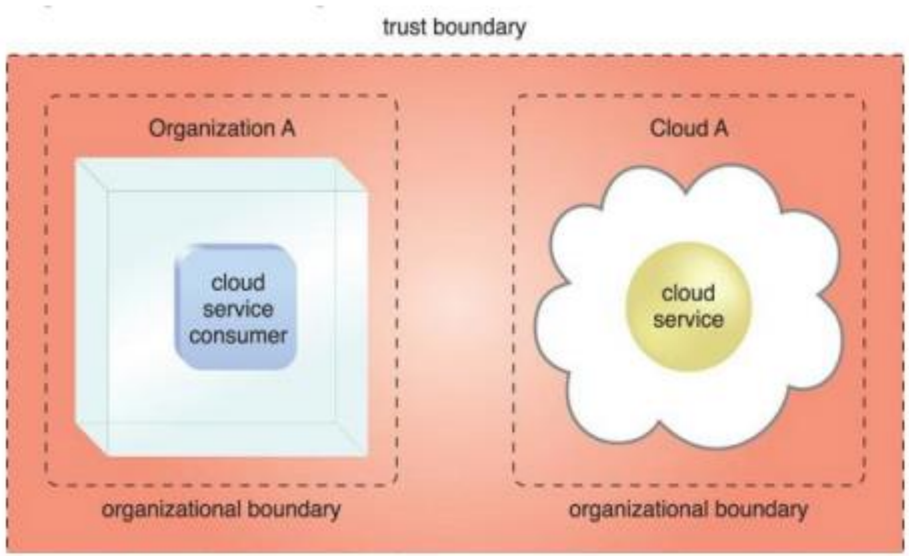


Figure 4.7. An extended trust boundary encompasses the organizational boundaries of the cloud provider and the cloud consumer.

CLOUD CHARACTERISTICS

on-demand usage

- A cloud consumer (like a person or company using the cloud) can set up and use cloud resources on their own, without needing help from the cloud provider.
- Once everything is set up, the resources can run automatically, without needing anyone to manage them all the time
- **"on-demand self-service"**, meaning you can use what you need, when you need it—**quickly and easily**.

ubiquitous access

- Ubiquitous access represents the ability for a cloud service to be widely accessible. Establishing ubiquitous access for a cloud service can require support for a range of devices, transport protocols, interfaces, and security technologies

Elasticity

- Elasticity means the cloud can automatically increase or decrease resources (like storage or computing power) based on what's needed at the time. It can scale up when demand is high and scale down when demand is low, without manual effort.

CLOUD CHARACTERISTICS

Measured usage

- Measured usage means the cloud can track how much and how long its resources (like storage, processing, or apps) are being used. This helps cloud providers: Charge only for what is used (like pay-as-you-go),Or simply monitor usage even if there's no cost (like in private clouds).It's not just for billing—it's also useful for monitoring performance, managing resources, and understanding how services are being used.

- **Resiliency**

Resilient computing means the cloud is set up to keep working even if something goes wrong. It does this by having backup copies of IT resources (like servers or apps) in different physical locations. If one fails, the system automatically switches to another one without stopping. This helps make sure that applications stay available and reliable, even during failures.

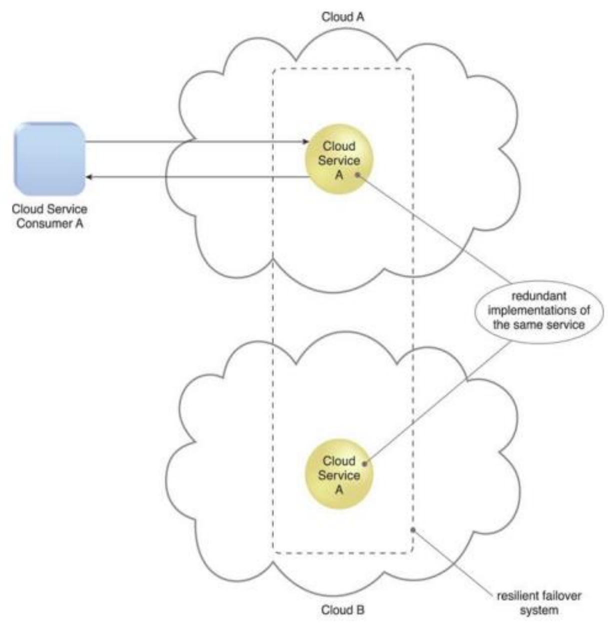


Figure 4.10. A resilient system in which Cloud B hosts a redundant implementation of Cloud Service A to provide failover in case Cloud Service A on Cloud A becomes unavailable.

CLOUD CHARACTERISTICS

multitenancy (and resource pooling)

- The characteristic of a software program that enables an instance of the program to serve different consumers (tenants) whereby each is isolated from the other, is referred to as multitenancy.
- Resource pooling allows cloud providers to pool large-scale IT resources to serve multiple cloud consumers.

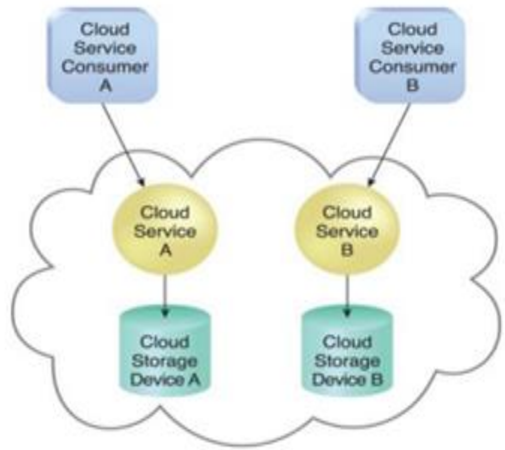


Figure 4.8. In a single-tenant environment, each cloud consumer has a separate IT resource instance.

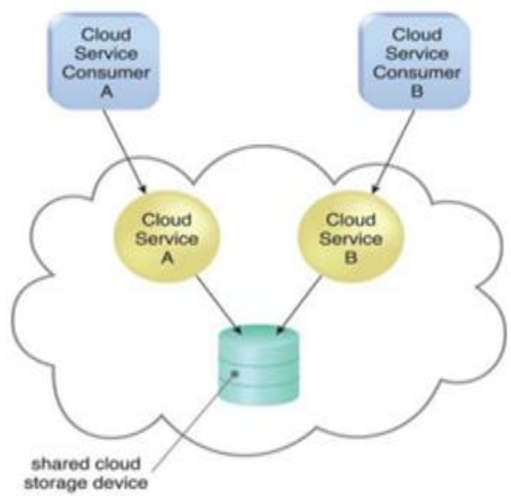


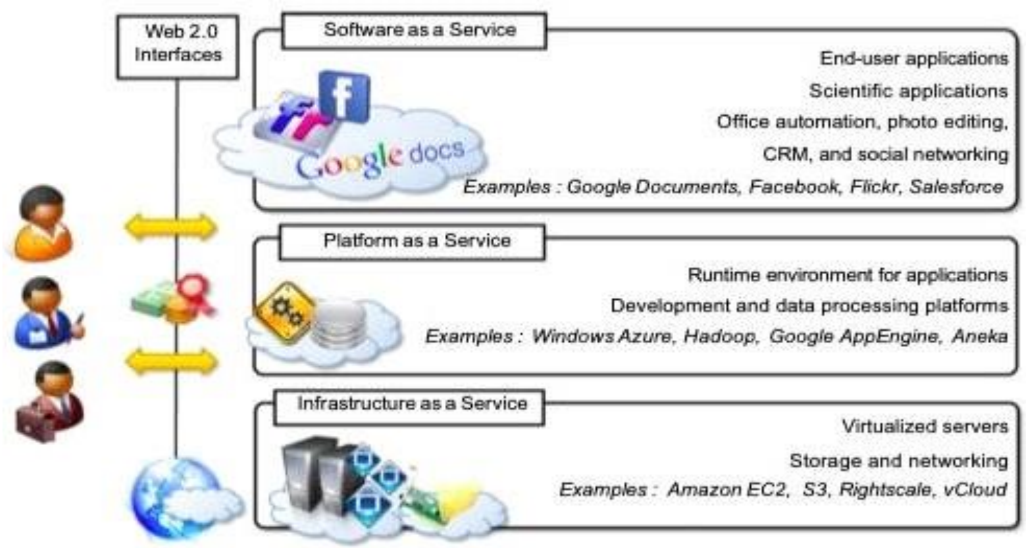
Figure 4.9. In a multitenant environment, a single instance of an IT resource, such as a cloud storage device, serves multiple consumers.

CLOUD DELIVERY MODELS

A cloud delivery model represents a specific, pre-packaged combination of IT resources offered by a cloud provider.

Three common cloud delivery models ARE:

- Infrastructure-as-a-Service (IaaS)
- Platform-as-a-Service (PaaS)
- Software-as-a-Service (SaaS)



INFRASTRUCTURE-AS-A-SERVICE (IAAS)

IaaS (Infrastructure as a Service) is a cloud model that gives users access to **basic IT resources** like servers, storage, networks, and operating systems over the internet.

You get **virtual versions of hardware**, not physical machines. You can **set it up the way you want**—just like building your own IT environment. It gives you **a lot of control and flexibility**, but you are also **responsible** for managing and configuring it.

The most common resource in IaaS is the **virtual server**, which you can customize (CPU, RAM, storage) based on your needs.

IaaS is great for tech-savvy users who want full control over their cloud setup.

Ex: Amazon Web Services (AWS), Microsoft Azure, Google Compute Engine.

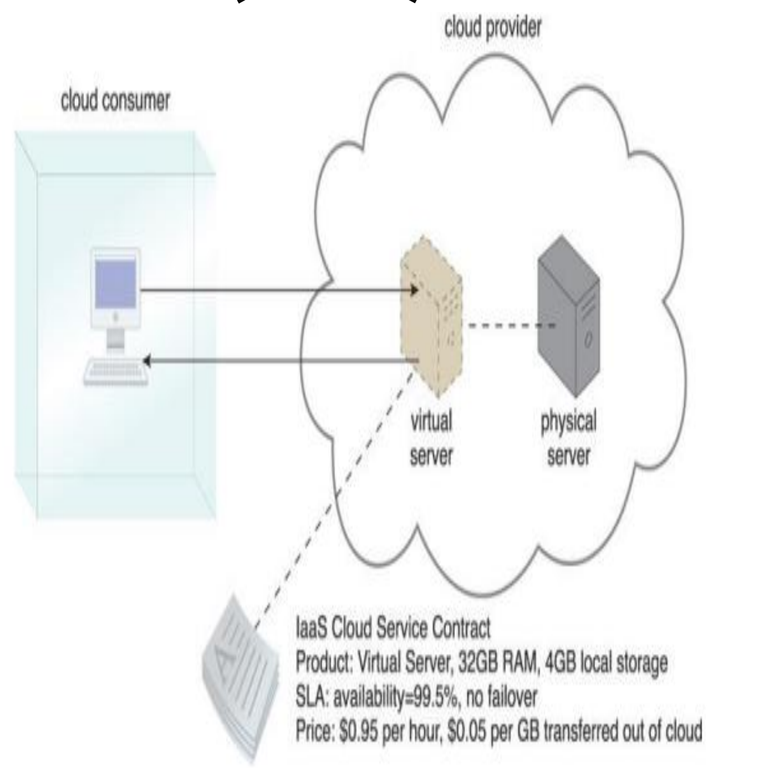


Figure 4.11. A cloud consumer is using a virtual server within an IaaS environment. Cloud consumers are provided with a range of contractual guarantees by the cloud provider, pertaining to characteristics such as capacity, performance, and availability.

PLATFORM-AS-A-SERVICE (PAAS)

PaaS gives users a ready-to-use platform with tools and resources to build, run, and manage applications (provides a runtime environment). It allows programmers to easily create, test, run, and deploy web applications. PaaS without having to set up the basic infrastructure.

It supports the full app development lifecycle (building, testing, deploying).

It is ideal for developers who want to focus on coding, not managing infrastructure.

PaaS includes infrastructure (servers, storage, and networking) and platform (middleware, development tools, database management systems, business intelligence, and more) to support the web application life cycle.

Ex: Google App Engine, Force.com, Joyent, Azure.

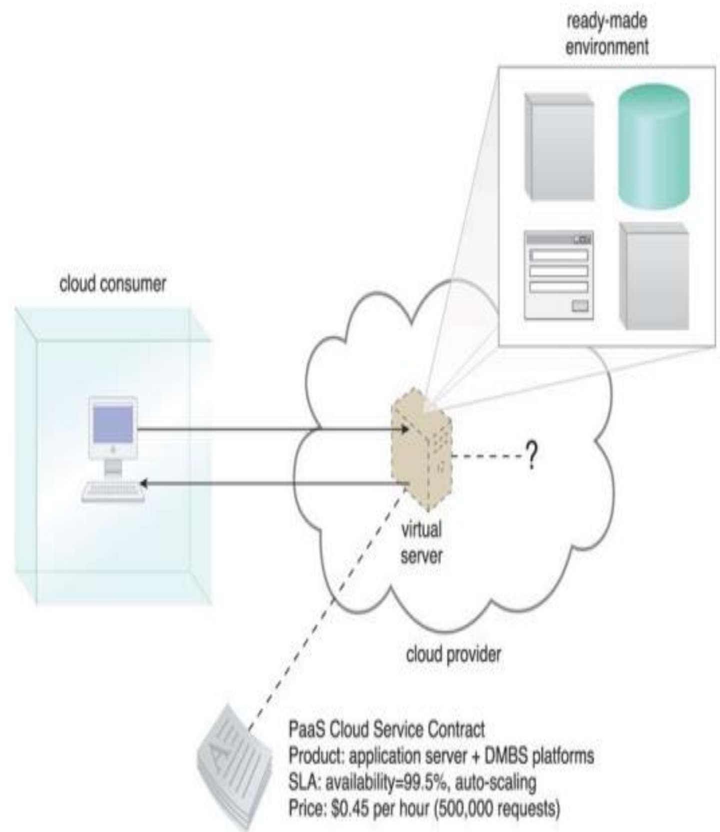


Figure 4.12. A cloud consumer is accessing a ready-made PaaS environment. The question mark indicates that the cloud consumer is intentionally shielded from the implementation details of the platform.

SOFTWARE-AS-A-SERVICE (SAAS)

SaaS is software provided as a **ready-to-use cloud service**, available to many users

SaaS is a shared software product accessed over the internet.

Users can use the software but can’t control or see the underlying systems.

It is easy to use, with no setup or maintenance required by the user.

Cloud providers manage everything; the user just logs in and uses the app.

Examples include Google Docs, Microsoft 365, Zoom, etc.

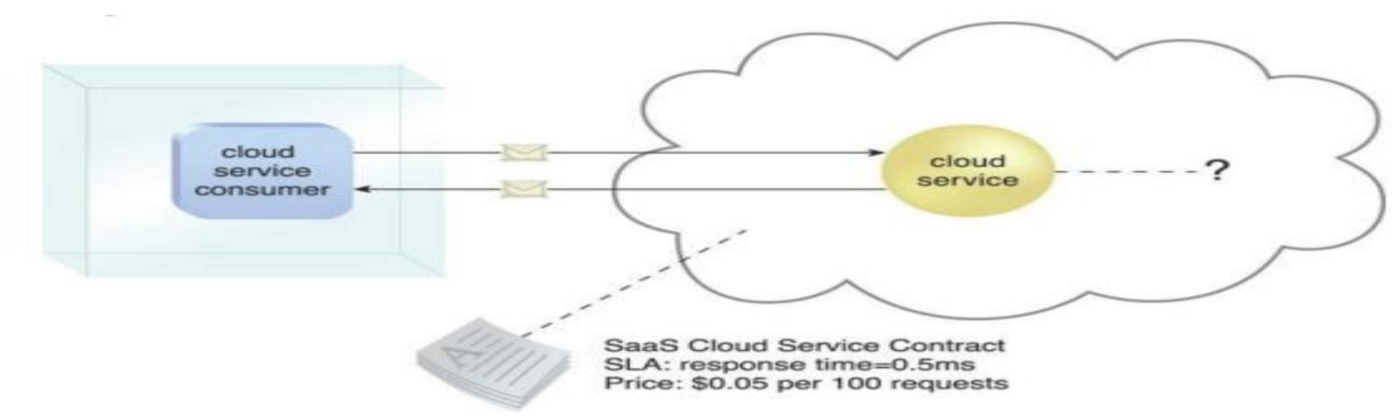
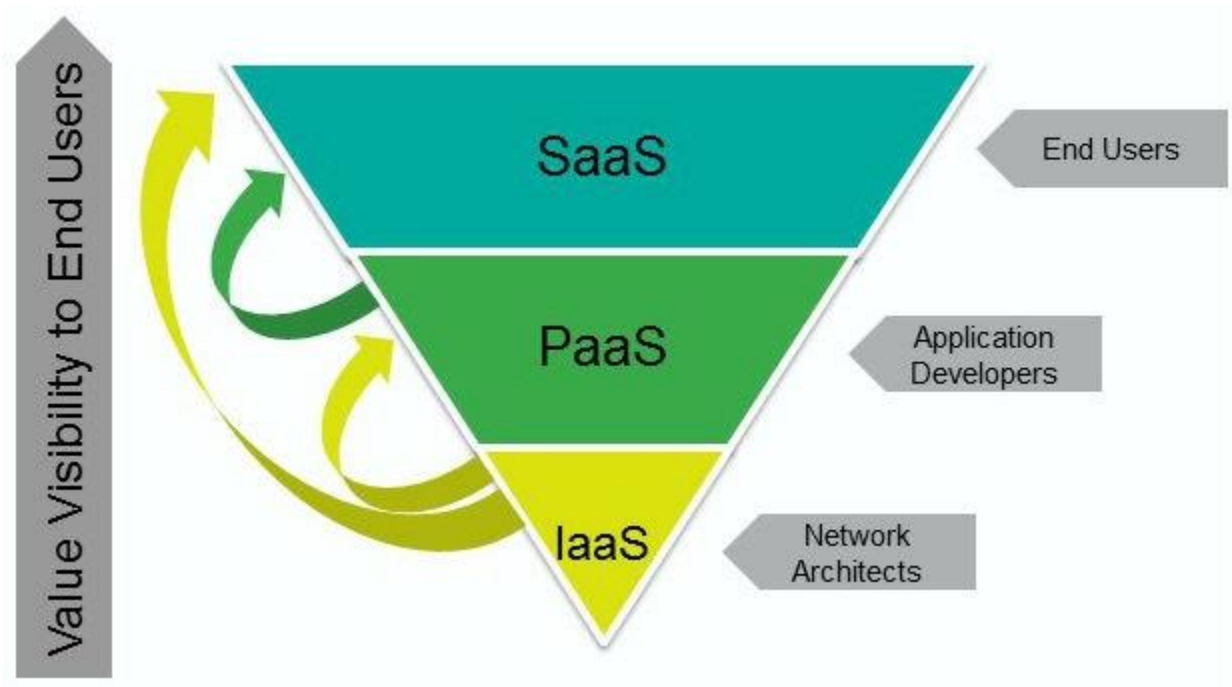


Figure 4.13. The cloud service consumer is given access the cloud service contract, but not to any underlying IT resources or implementation details.

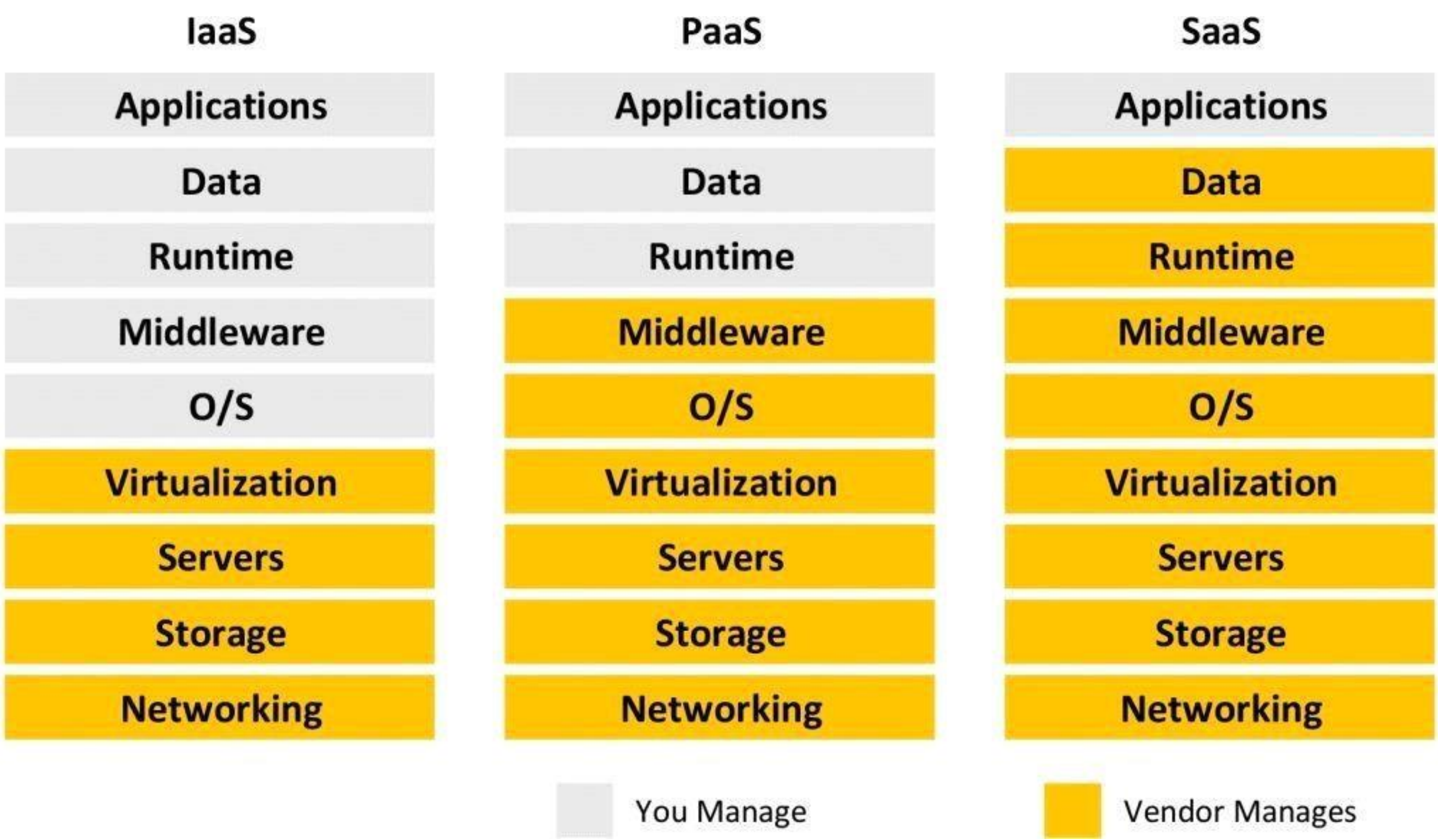
CLOUD DELIVERY MODELS



Cloud Delivery Model	Typical Level of Control Granted to Cloud Consumer	Typical Functionality Made Available to Cloud Consumer
SaaS	usage and usage-related configuration	access to front-end user-interface
PaaS	limited administrative	moderate level of administrative control over IT resources relevant to cloud consumer's usage of platform
IaaS	full administrative	full access to virtualized infra-structure-related IT resources and, possibly, to underlying physical IT resources

Cloud Delivery Model	Common Cloud Consumer Activities	Common Cloud Provider Activities
SaaS	uses and configures cloud service	implements, manages, and maintains cloud service monitors usage by cloud consumers
PaaS	develops, tests, deploys, and manages cloud services and cloud-based solutions	pre-configures platform and provi-sions underlying infrastructure, middleware, and other needed IT resources, as necessary monitors usage by cloud consumers
IaaS	sets up and configures bare infrastructure, and installs, manages, and monitors any needed software	provisions and manages the physical processing, storage, networking, and hosting required monitors usage by cloud consumers

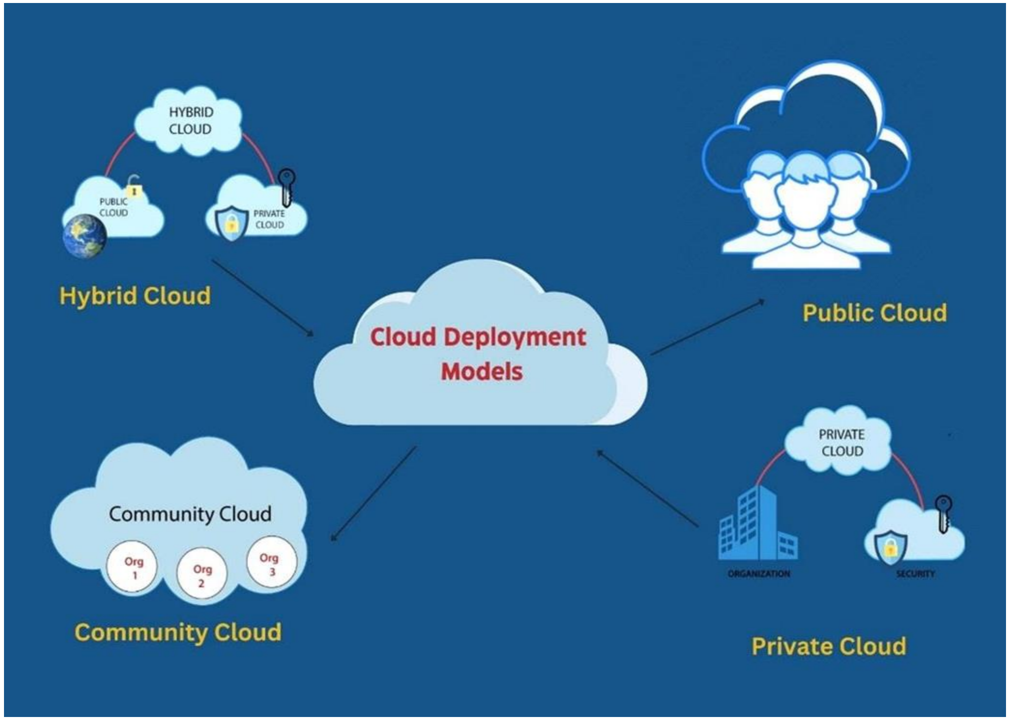
COMPARING CLOUD DELIVERY MODELS



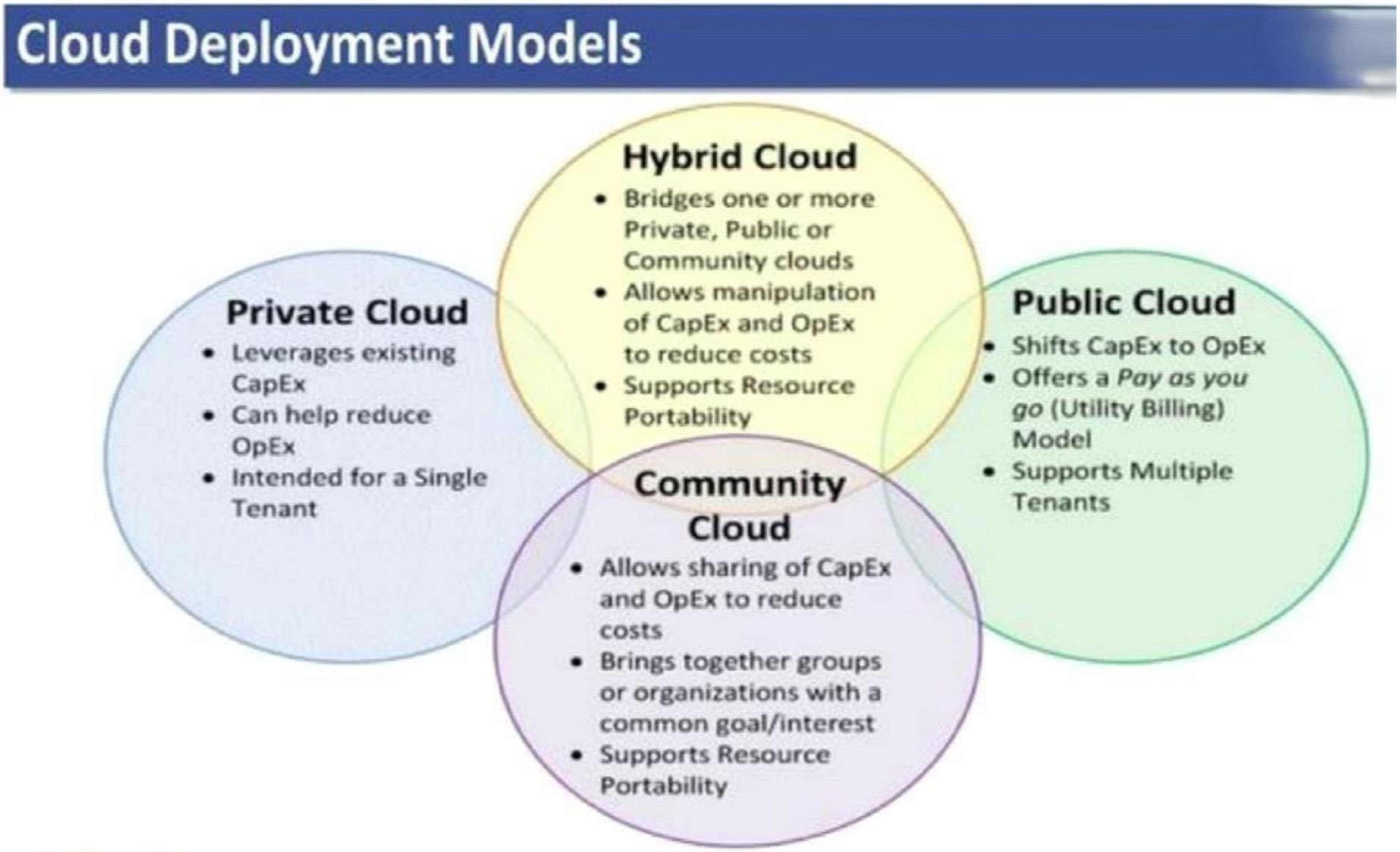
CLOUD DEPLOYMENT MODELS

A cloud deployment model represents a specific type of cloud environment, primarily distinguished by ownership, size, and access. There are four common cloud deployment models:

- Public cloud
- Community cloud
- Private cloud
- Hybrid cloud



CLOUD DEPLOYMENT MODELS



PUBLIC CLOUD

- A public cloud is a publicly accessible cloud environment owned by a third-party cloud provider.
- The cloud provider is responsible for the creation and on-going maintenance of the public cloud.
- The public cloud is one in which cloud infrastructure services are provided over the internet to the general people or major industry groups.
- Pros: Minimal Investment, No setup cost, Infrastructure Management is not required, No maintenance and Dynamic Scalability.
- Cons: Less secure ,Low customization

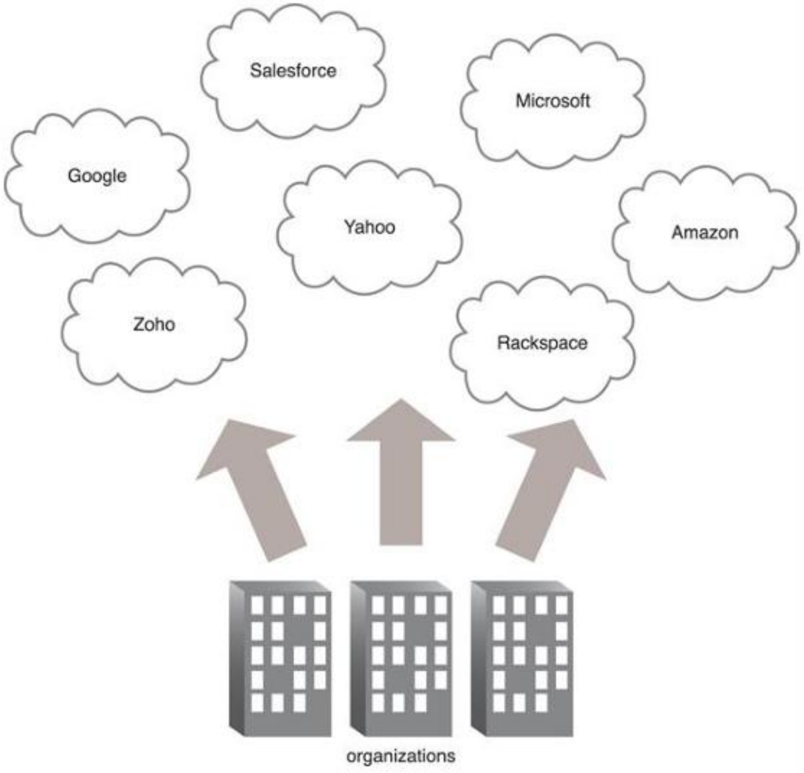


Figure 4.17. Organizations act as cloud consumers when accessing cloud services and IT resources made available by different cloud providers.

PRIVATE CLOUDS

A private cloud is owned by a single organization. Private clouds enable an organization to use cloud computing technology as a means of centralizing access to IT resources by different parts, locations, or departments of the organization.

It’s a one-on-one environment for a single user (customer)

The cloud platform is implemented in a cloud-based secure environment that is protected by powerful firewalls and under the supervision of an organization’s IT department. The private cloud gives greater flexibility of control over cloud resources

Pros: Better Control, Data Security and Privacy, Customization

Cons: Costly, Less scalable- Private clouds are scaled within a certain range as there is less number of clients.

PRIVATE CLOUDS

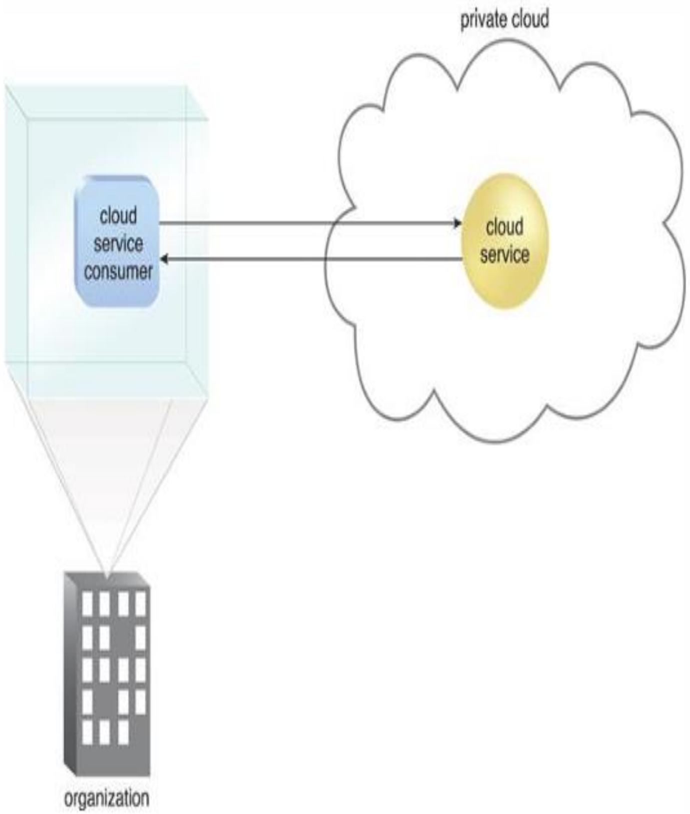
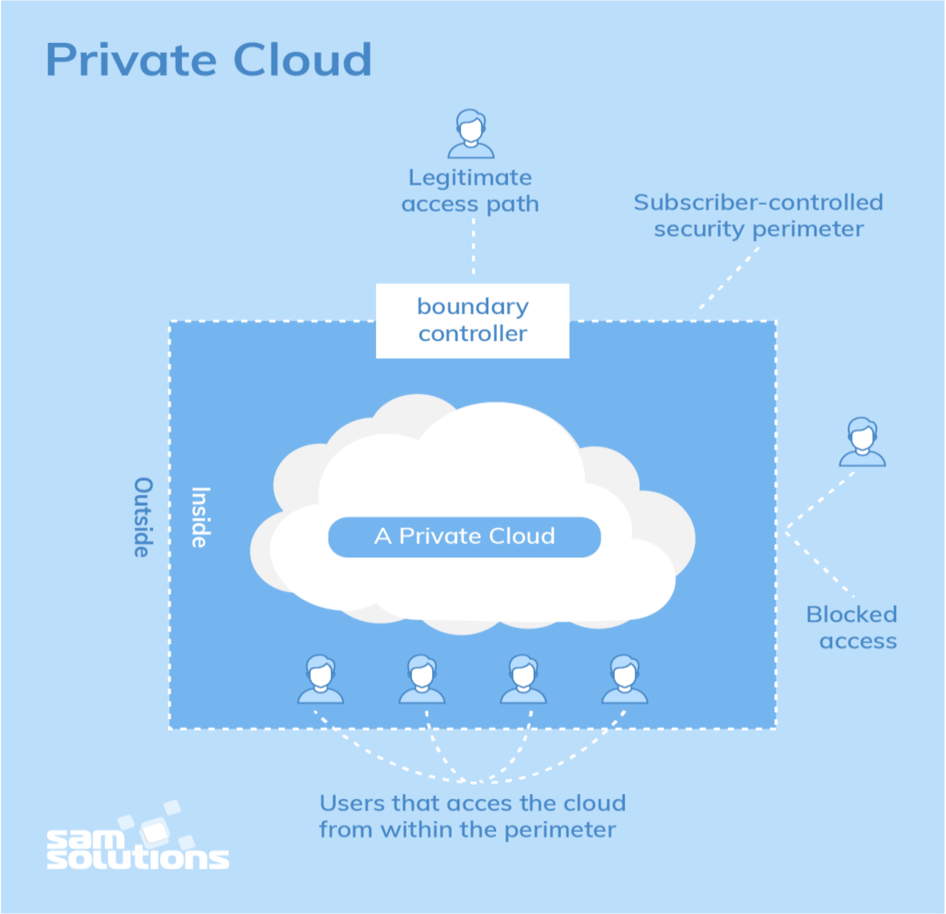


Figure 4.19. A cloud service consumer in the organization's on-premise environment accesses a cloud service hosted on the same organization's private cloud via a virtual private network.



COMMUNITY CLOUDS

A community cloud is similar to a public cloud except that its access is limited to a specific community of cloud consumers. The community cloud may be jointly owned by the community members or by a third party cloud provider that provisions a public cloud with limited access. The member cloud consumers of the community typically share the responsibility for defining and evolving the community cloud

The infrastructure of the community could be shared between the organization which has shared concerns or tasks. It is generally managed by a third party or by the combination of one or more organizations in the community.

Pros: Cost Effective, Security, Shared resources, Collaboration and data sharing:

Cons: Limited Scalability, Rigid in customization.

COMMUNITY CLOUDS

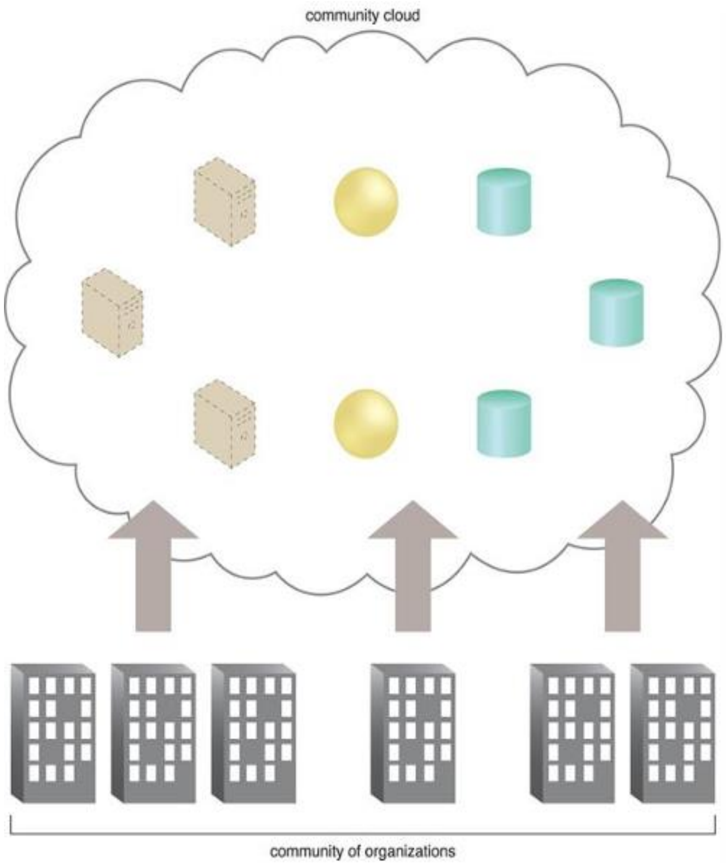
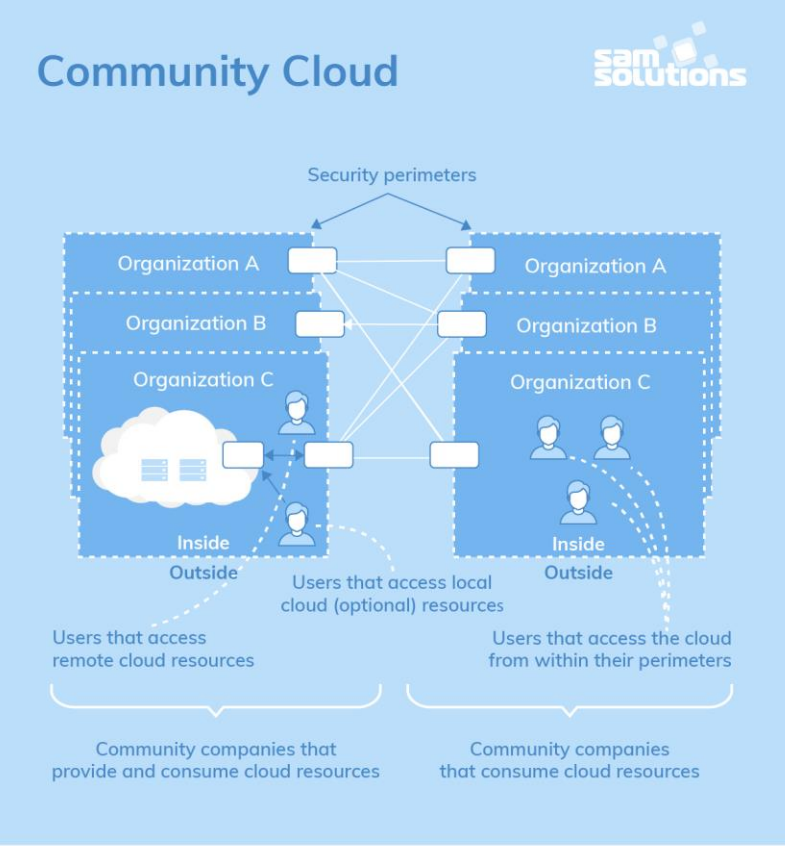


Figure 4.18. An example of a "community" of organizations accessing IT resources from a community cloud.



HYBRID CLOUD

Hybrid cloud is any environment that uses both public and private clouds. By bridging the public and private worlds with a layer of proprietary software, hybrid cloud computing gives the best of both worlds

A hybrid cloud is a cloud environment comprised of two or more different cloud deployment models. For example, a cloud consumer may choose to deploy cloud services processing sensitive data to a private cloud and other, less sensitive cloud services to a public cloud. The result of this combination is a hybrid deployment model.

Pros: Flexibility and control, Security, Reasonable price

Cons: Difficult to manage, Slow data transmission.

HYBRID CLOUD

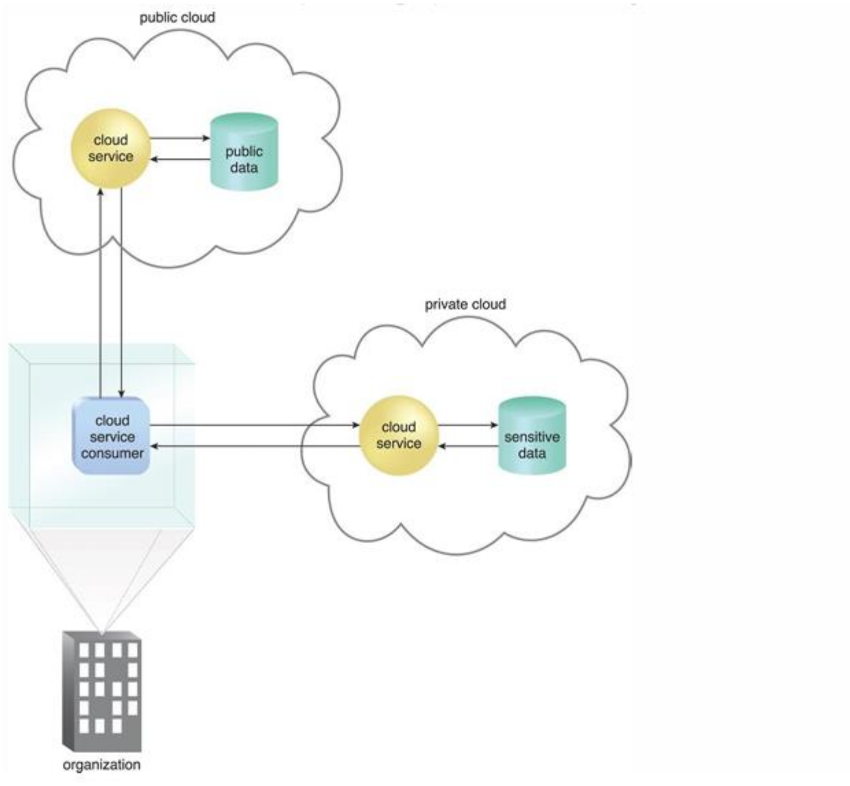


Figure 4.20. An organization using a hybrid cloud architecture that utilizes both a private and public cloud.

CLOUD DEPLOYMENT MODELS

	Public	Private	Community	Hybrid
Ease of setup and use	Easy	Requires IT proficiency	Requires IT proficiency	Requires IT proficiency
Data security and privacy	Low	High	Comparatively high	High
Data control	Little to none	High	Comparatively high	Comparatively high
Reliability	Low	High	Comparatively high	High
Scalability and flexibility	High	High	Fixed capacity	High
Cost-effectiveness	The cheapest	Cost-intensive; the most expensive model	Cost is shared among community members	Cheaper than a private model but more costly than a public one
Demand for in-house hardware	No	Depends	Depends	Depends